Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/08/2004
PI Name:	Mazuruk, Konstantin Ph.D.		
Project Title:	Effects of Traveling Magnetic Field on Dynamics of Solidification		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	07/01/2000	**End Date:**	09/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Grugel, Richard (NASA Marshall Space Flight Center)		
Grant/Contract No.:	NAS8-00128		
Performance Goal No.:			
Performance Goal Text:			
	Traveling Magnetic Field (TMF) is based on imposing a controlled phase-shift in a train of electromagnets, forming a stack. TMF is based on imposing a controlled phase-shift in a train of electromagnets, forming a stack. Thus, the induced magnetic field can be considered to be traveling along the axis of the stack. The coupling of this traveling wave with an electrically conducting fluid results in a basic flow in a form of a single axisymmetric roll. The magnitude and direction of this flow can be remotely controlled. Furthermore, it is possible to localize the effect of this force field though activating only a number of the magnets. This force field generated in the fluid can, in principle, be used to control and modify convection in the molten material. For example, it can be used to enhance convective mixing in the melt, and thereby modify the interface shape, and macrosegregation. Alternatively, it can be used to counteract thermal and/or solutal buoyancy forces. High frequency TMF can be used in containerless processing techniques, such as float zoning, to affect the very edge of the fluid so that Marangoni flow can be counter balanced.		

Task Description:	The proposed program consists of basic fundamentals and applications. Our goal in conducting the following experiments and analyses is to establish the validity of TMF as a new tool for solidification processes. Due to its low power consumption and simplicity of design, this tool may find wide spread use in a variety of space experiments. The proposed ground based experiments are intended to establish the advantages and limitations of employing this technique. In the fundamentals component of the proposed program, we will use theoretical tools and experiments with mercury to establish the fundamental aspects of TMF-induced convection through a detailed comparison of theoretical predictions and experimental measurements of flow field. In this work, we will conduct a detailed parametric study involving the effects of magnetic field strength, frequency, wave vector, and the fluid geometry. The applications component of this work will be focused on investigating the effect of TMF on the following solidification and pre-directional solidification processes: 1) Bridgman growth of Ga:Ge with the goal of counteracting the buoyancy-driven convection; 2) Mixing of Pb-Ga and Pb-Sn alloys with the aim of initiating and maintaining a uniform melt prior to solidification processing; and 3) Float Zone growth with the aim of identifying, through simulations and model experiments, conditions needed to counteract Marangoni flow in a microgravity environment. 1) to develop theoretical and experimental fundamental features of the traveling magnetic field (TMF) technique, with the aim of identifying its benefits for microgravity research on solidification processing 2) to determine the influence of a traveling magnetic field on the following selected technologies: a) crystal growth of semiconductor materials by Bridgman and Float Zone techniques mixing of metal alloy melts prior to solidification The proposed research has strong relevance to microgravity research and the objectives of the NRA. TMF can provide a unique and accurate mechanism for generation and control of desirable flow patterns for microgravity research. These attributes have significant relevance to 1) Alloy mixing prior solidification in a microgravity environment. TMF can provide this mixing with a low level of power consumption; 2) TMF can offset the deleterious effects of Marangoni convection in microgravity containerless processing. Thus, TMF can be instrumental in further understanding this phenomena; 3) Generation of controlled flows will allow the investigation of the effect of these flows on growth morphology and growth kinetics; and 4) On Earth, TMF has the potential to significantly counter-balance thermosolutal convection, thereby creating conditions similar to those obtained in microgravity. Once demonstrated, this new tool for use in solidification has the strong potential to find applications in a host of microgravity material research projects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The primary effort has been focused on developing a deeper understanding of fluid dynamics phenomena. Here, for a pipe geometry, the flow induced by TMF was found to be marginally stable. Although it is known that pipe flow is stable, no study of the case with a volumetric body force exists. A linear stability analysis was done for both unidirectional and recirculating pipe flow. Neither case revealed a critical point. However, for confined volumes, an axisymmetric basic flow is unstable at some magnetic Taylor number. This Hopf bifurcation shows a typical hysteresis behavior. The phenomenon was quantified and the critical Taylor number as a function of the aspect ratio was determined. This research reinforces the previous experimental finding that the onset of instability for both TMF and RMF occurs for the same range of magnetic Taylor numbers. This work is in preparation for publication. The non-axisymmetric mode stability analysis is also in progress. Crystal growth modeling has again been refined. In essence, it reinforces that TMF can reduce natural convective flow by as much as 20 times in a standard Vertical Bridgman crystal growth configuration. This work has been published. The study of the Lorentz force for arbitrary frequencies and for arbitrary aspect ratios has been completed. Among a number of features, an induced volumetric body force changes its sign in the vicinity of one of the end cups, a rather counterintuitive effect. This work contributes to the fundamentals of the TMF and has been published.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Yesilyurt S, Motakef S, Grugel R, Mazuruk K. "The effect of the traveling magnetic field (TMF) on the buoyancy-induced convection in the vertical Bridgman growth of semiconductors." Journal of Crystal Growth Vol. 263 (2004)80-89. https://doi.org/10.1016/j.jcrysgro.2003.11.066 , Mar-2004
Articles in Peer-reviewed Journals	Volz MP, Mazuruk K. "Lorentz body force induced by travelling magnetic fields." Magnetohydrodynamics. May 2004; 40(2):117-126. , May-2004